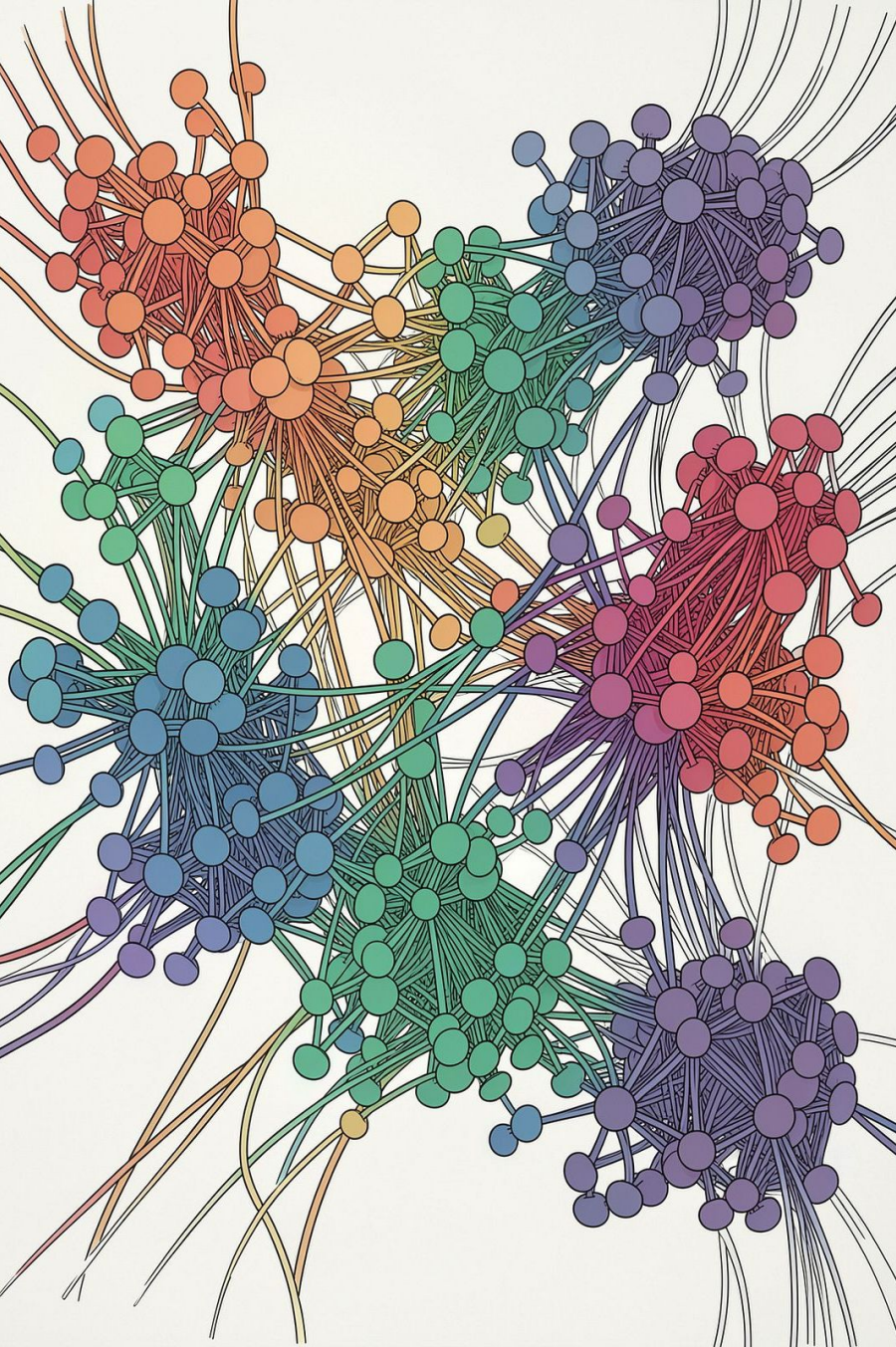




Malicious Traffic Pattern Clustering Using K-Means and Hierarchical Clustering

Pakkiru Sai Harshith Reddy – 2520030375

Kasula Yashas – 2520030401

Machine Learning – Department of Computer Science and Engineering

Under the Guidance of: Mr. Shaik Asif, Assistant Professor, Department of Computer Science and Engineering

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Roadmap with clear milestones – each section focuses on rigorous analysis, reproducible experiments and interpretable results.

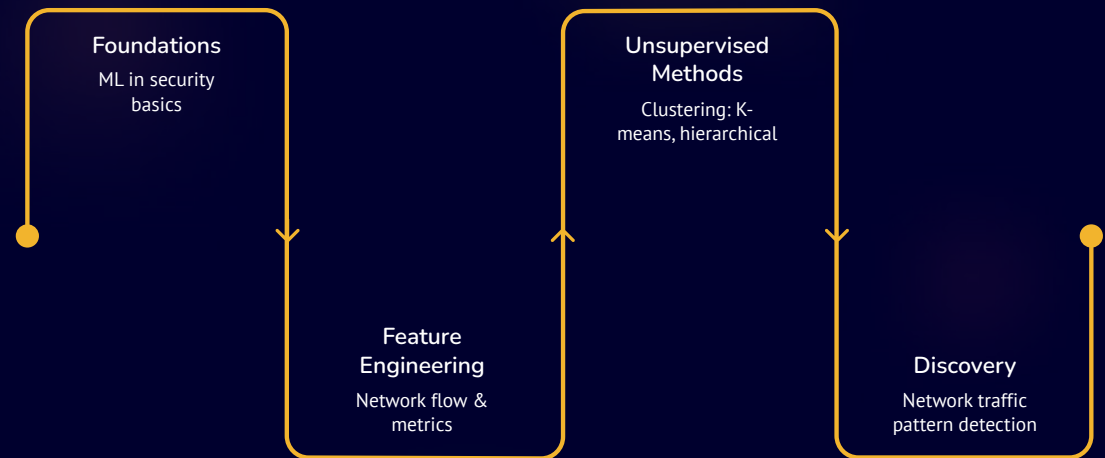
Literature

- Machine Learning in Cybersecurity
- Unsupervised Learning
- Network Traffic Analysis
- Clustering for Intrusion Detection
- K-Means Clustering
- Hierarchical Clustering
- CICIDS2017



Research Direction: Using unsupervised clustering to discover meaningful patterns within network traffic.

Note: Specific citations will be added later as placeholders for primary papers and datasets.





Problem Statement

Modern networks generate enormous volumes of traffic, making suspicious behaviour difficult to identify manually.

Massive Network Traffic

Complex Traffic Behaviour

Hidden Patterns

Difficult Manual Analysis

Conclusion: Our project explores unsupervised clustering to discover meaningful traffic patterns and support intelligent analysis.

Abstract

As network volumes and protocol complexity increase, detecting malicious activity through manual inspection becomes infeasible. This project applies unsupervised learning to CICIDS2017 network-flow data to surface latent traffic structures. The pipeline includes careful preprocessing, feature selection, and dimensionality reduction. We apply K-Means and Hierarchical Clustering, visualise clusters with PCA, and evaluate separation using silhouette analysis. The final system will support an interactive dashboard for analysts to inspect suspicious clusters and interpret behavioural traits.



Data Analysis

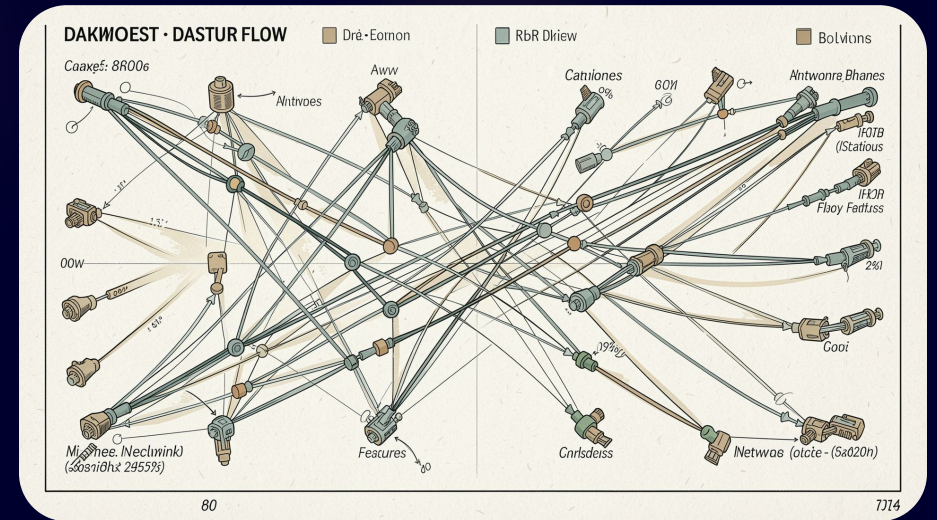
Dataset: CICIDS2017 – CSV format, network-flow granularity.

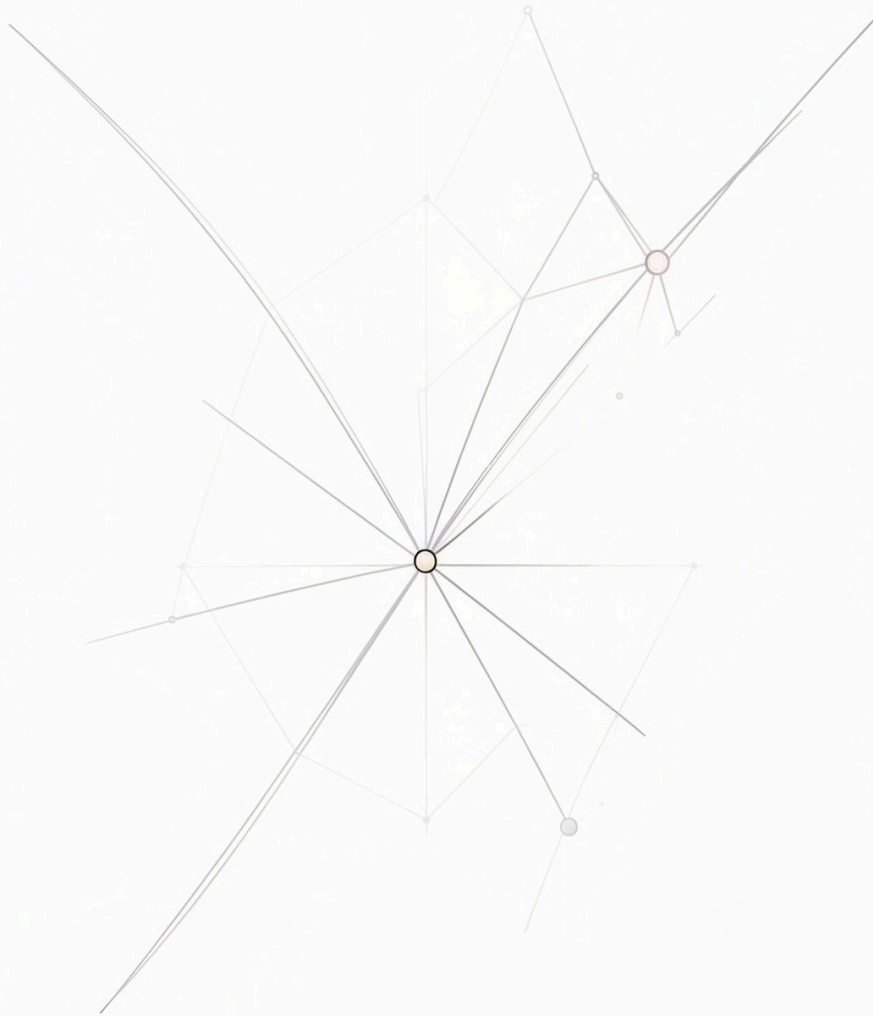
Key properties:

- Network-flow based traffic records
- Contains benign and malicious traffic
- Multiple attack categories including volumetric and application-layer attacks
- Rich set of network-flow features for behavioural characterisation

Representative categories: BENIGN, DDoS, PortScan, DoS, Brute Force, Web Attacks, Infiltration, Bot, Heartbleed

Representative features: Flow Duration, Packet Count, Packet Length,
Flow Bytes/s, Flow Packets/s





Methodology

Content to be Added



Implementation & Results

Content to be Added

Reserved area for: Dashboard screenshots, K-Means results, Hierarchical clustering results, PCA visualisation, Silhouette Score, Algorithm comparison, Cluster analysis.

Conclusion

1 Unsupervised value

Unsupervised clustering can reveal hidden structures within network traffic.

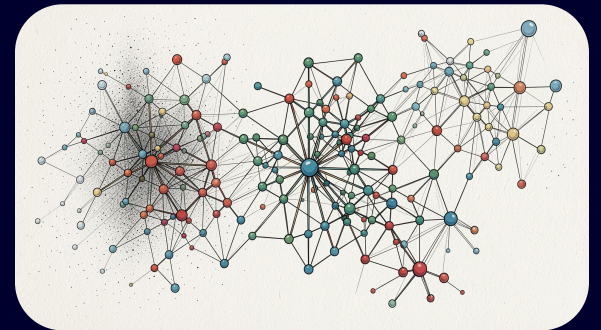
2 Complementary methods

K-Means and Hierarchical Clustering provide complementary perspectives for traffic analysis – partitioning and nested structure.

3 System goal

The proposed system aims to support analysts by identifying and interpreting suspicious traffic patterns.

From raw network traffic to meaningful behavioral patterns.



Thank You Questions & Discussion

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